

<!DOCTYPE html>

ETSI GS QKD 004 push mode specification, QKD server

etsi004

postopen key stream

getstart pushed key retrieval process

putclose the key stream

API docs by Redocly

ETSI GS QKD 004 push mode specification, QKD server (1.0.0)

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URL: <https://qkd-kms.ait.ac.at>

Server description of the QKD endpoint processing ETSI GS QKD 004 push mode requests

etsi004

open key stream

The open-connect message connects a KMS to the QKD Device. Conforming to the specification, the QKD device will assign the `key_stream_id` and can suggest alternative QoS due to its capabilities.

Request Body schema: application/json

source

required

string <uri>

the source URI of the request.

destination

required

string <uri>

the destination URI of the request.

key_stream_id

string <uuid>

key stream ID. If not specified it is up to the QKD device to define the UUID

qos

required

object (qos)

requested QoS

Responses

201

Key stream successfully created

400

Client sent a malformed input

500

The server experienced an internal error

post/open-connect

https://qkd_server/api/v1/kms/etsi004/open-connect

Request samples

Payload

Content type

application/json

Copy

Expand all

Collapse all

```
{
  "source": "http://kms-1.net",
  "destination": "http://kms-2.net",
  "key_stream_id": "4eb81a5c-031f-4d1f-881d-309bee44fc20",
  "qos": {
    "key_chunk_size": 32,
    "max_bps": 128,
    "min_bps": 64,
    "jitter": 1,
    "priority": 3,
    "timeout": 50,
    "ttl": 600,
    "metadata_mimetype": "application/json"
  }
}
```

Response samples

201

Content type

application/json

Copy

Expand all

Collapse all

```
{
  "source": "http://kms-1.net",
  "destination": "http://kms-2.net",
  "key_stream_id": "4eb81a5c-031f-4d1f-881d-309bee44fc20",
  "qos": {
    "key_chunk_size": 32,
    "max_bps": 128,
    "min_bps": 64,
    "jitter": 1,
    "priority": 3,
    "timeout": 50,
    "ttl": 600,
    "metadata_mimetype": "application/json"
  }
}
```

}

“status”: 0

}

start pushed key retrieval process

The get-key message is the starting point after which the QKD device can start pushing keys.

Request Body schema: application/json

key_stream_id

required

string <uuid>

corresponding to the one given in open_connect

index

integer <uint32>

increasing index of the key chunk in the key stream. Wrapping at UINT_MAX.

metadata

object (metadata)

Responses

205

Successful registration

400

Client sent a malformed input

500

The server experienced an internal error

get/get-key

https://qkd_server/api/v1/kms/etsi004/get-key

Request samples

Payload

Content type

application/json

Copy

Expand all

Collapse all

{

“key_stream_id”: “4eb81a5c-031f-4d1f-881d-309bee44fc20”,

“index”: 0,

“metadata”: {

“size”: 0,

“buffer”: ””

}

}

Response samples

205

Content type

application/json

Copy

Expand all

Collapse all

```
{
  "status": 0,
  "metadata": {
    "size": 0,
    "buffer": ""
  }
}
```

close the key stream

The close message disconnects the KMS from the QKD device.

Request Body schema: application/json

key_stream_id

required

string <uuid>

corresponding to the one given in open_connect

Responses

204

Key stream successfully closed

400

Client sent a malformed input

500

The server experienced an internal error

put/close

https://qkd_server/api/v1/kms/etsi004/close

Request samples

Payload

Content type

application/json

Copy

```
{
  "key_stream_id": "4eb81a5c-031f-4d1f-881d-309bee44fc20"
}
```

Response samples

204

Content type

application/json

Copy

```
{  
  "status": 0  
}
```